Dimethyl Methylphosphonate Responds Opposite to Soman, Cyclosarin and VX in a Colorimetric Array of Authentic Chemical Warfare Agents

Ku Kang,[†] Jeongyun Kim,[†] Miae Kim,[†] Doo-Hee Lee,[‡] Jin Yoo,[†] and Yoon Jeong Jang[*,†]

†*Chemical, Biological, Radiological and Nuclear (CBRN) Defense Research Institute, Ministry of National Defense, Seoul, Republic of Korea*

‡*Department of Civil Engineering and Environmental Sciences, Korea Military Academy, Seoul 01805, Republic of Korea*

E-mail: yjjangchem@korea.kr

Abstract

Chemical warfare agent detection is qualified on simulants, and we asked what measurement establishes that one reproduces its agent. We reanalyse the colorimetric tables of three published runs of one 29-component array, which measured sixteen authentic agents at four concentrations in two optical channels. Dimethyl methylphosphonate, the standard nerve agent simulant, gives a response pattern positively correlated with sarin, $+0.65$ and $+0.57$ in the two channels, and anti-correlated with soman, -0.49 and -0.48, with cyclosarin, -0.72 and -0.49, and with VX, -0.70 and -0.27; tabun changes sign between channels. The two channels are read from the same wells and are not independent replicates, the comparison comes from one run, and the tables carry no

repeats, so we report this as a measurement and not as a test. The array does discriminate on average: in all six run-channel cells every compound agrees with itself across concentrations more closely than with the mean of the other compounds, and all six clear $q = 0.05$ against a label-permutation null. Descriptor distance, the basis on which simulants are selected, does not resolve the difference. Twelve exact Mantel tests and six combinations of them leave nothing below $q = 0.05$, but the design resolves a rank correlation of 0.85 at 80% power at the corrected threshold and the largest observed is 0.57, so that is a limit on the design, not a demonstration of absence. Across the three runs, one designated simulant has been measured beside the agents.

Introduction

Nearly every chemical warfare agent detector in the literature is developed, optimised and qualified on compounds that are not chemical warfare agents.[1,2] Dimethyl methylphosphonate stands in for the G-series nerve agents,[3] dipropylene glycol monomethyl ether for training detectors,[4] and methyl salicylate for vapour penetration through protective ensembles.[5,6] The substitution is unavoidable, because the agents are lethal at concentrations where handling them requires a licensed facility, and it is load-bearing: a sensor reported as selective for sarin has, in most cases, never been exposed to sarin.

What licenses the substitution is a claim of similarity. The compound chosen is said to be agent-like, and similarity is read from molecular descriptors: vapour pressure, dipole moment, frontier orbital energies, substructure and shape. We test the first of those separately, on measured vapour pressures, in Section . Cheminformatic selection on exactly those grounds has been proposed as the systematic way to find a simulant,[7] reviews catalogue the compounds in use for environmental behaviour,[8] for emergency decontamination trials[9] and for toxic industrial chemicals,[10] and the US Army maintains a test operations procedure governing simulant selection.[11] The claim that matters for a detector is different: that the substitute produces the response the agent would produce in the same device. The

two claims are connected only if descriptor similarity transfers to response similarity.

That transfer has been tested once, for a different measurement and in simulation: Agrawal and co-workers computed dilute-limit adsorption for four common simulants and for sarin and soman across 2969 metal–organic frameworks under two force fields.[12] That answers the question for adsorption, not for detection, because a detector response is not an adsorption isotherm and a simulation of the agent is not a measurement of it. The measured comparison has not been made, because making it requires the agents, and response data on authentic agents exist in few places where the compounds used as simulants are run alongside them. A recent survey of threat simulants records that they have generally been selected on the basis of their historical use, and describes no governing document on the chemical warfare agent side.[13] A critical review of the low-toxicity simulants in current use states the criterion plainly, as alignment of physicochemical properties such as hydrophobicity, volatility and solubility with those of the agent.[14] That is a statement about the compound and not about the detector, and nothing in the review measures whether a compound selected that way produces the response the agent produces.

Asking whether the components a detection claim rests on have themselves been validated is a question this field has begun to put in print, most recently for aptamers.[15] Simulants are the analyte-side form of that problem, and the concern has been stated in this journal: reporting a bioelectronic nose tested on authentic sarin and VX, two of us and our co-workers noted that detection research built on structural simulants may not carry over to the agents, depending on the selectivity of the sensor.[16] That work asserted the risk and worked around it by measuring the agents, as did our group's field-deployable detection work.[17] Neither quantified it, which requires putting the descriptor distances and the measured responses side by side and asking whether one predicts the other. The contribution here is that measurement and not the assertion. The three source studies were run at an OPCW designated laboratory, which identifies and assays authentic agents under an accredited procedure; the reanalysis reported here handled no agent and made no measurement of its own.

This work reports that measurement. A 29-component colorimetric array recorded the responses of sixteen authentic agents, twelve of them Chemical Weapons Convention Schedule 1 compounds and four Schedule 3 industrial chemicals, at four solution concentrations in two optical channels. The result that bears most directly on practice is the simplest one in the paper: dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, responds opposite to soman, cyclosarin and VX in both optical channels and tracks only sarin. Descriptor distance, the basis on which that substitution was made, carries no information about response distance that this design can resolve. Where the designation record and the regulatory toxicity axis might have supplied the missing evidence, the record names measurements that are not detection and the axis the agents are placed on does not reach the incumbents.

This is not a new sensing method, and we propose no replacement compound. It is a measurement of whether the substitutes the field already relies on carry the property they are relied upon for.

Results and Discussion

The array separates the agents

A comparison between descriptor space and response space is worth making only if the response measurement resolves the compounds at all, and that question is already answered in the reports these data come from. On the first run a linear discriminant classifier on a six-dye subset under dual illumination separated the four G agents, VX and dimethyl methylphosphonate at 100% accuracy, against 87.5% under visible illumination alone.[18] On the second, a six-dye subset classified all seven agents at an accuracy of 0.964 with a macro F_1 of 0.964 under five-fold stratified cross-validation.[19] On the third, a four-dye subset separated the four Novichok agents at an accuracy of 1.000 and a macro F_1 of 1.000 under leave-one-out cross-validation, over a working range of 10 to 500 μM with a detection limit near 10 μM.[20]

Three qualifications move where the premise rests. Each figure is obtained on a dye subset chosen for that run, not on the 29-component array we analyse; the first run reports no held-out evaluation, so its 100% is an apparent rather than a validated accuracy; and that first run is the only one in which a designated simulant was measured beside the agents, so the two cross-validated figures come from runs that contain no simulant. Neither cross-validated figure can be reproduced from the deposited tables, which are averaged over replicates, so we quote them without verifying them. The premise for the first run therefore rests on our own check below, over all six run-channel cells.

Our own check is narrower, and we report it because the comparison that follows is made on patterns rather than on class labels. Each exposure gives a 29-dye ΔE_{00} vector, which we scaled to unit length, centred dye-wise at each concentration and scaled to unit length again, so that agreement, taken as the cosine, is measured between patterns and not magnitudes. It answers whether a compound's pattern is stable enough across dose to be treated as that compound's pattern at all.

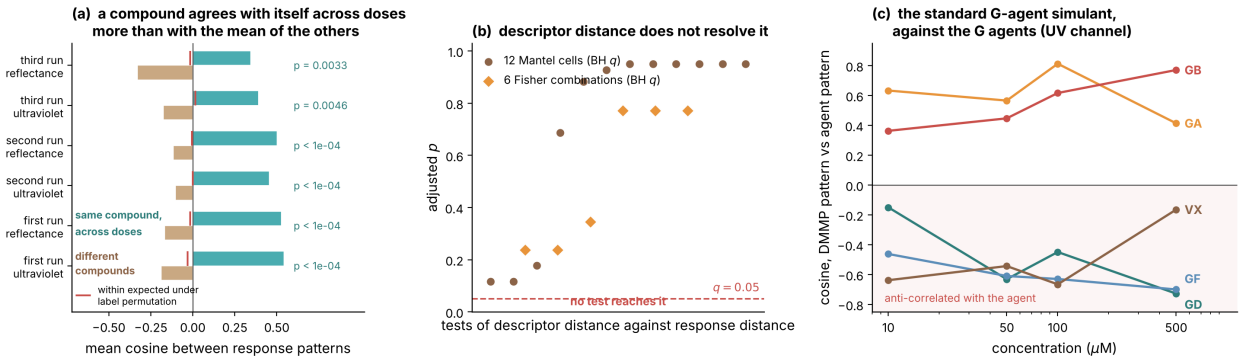


Figure 1: Response-space tests on authentic agents. (a) Mean cosine between response patterns, within one compound across concentrations and between different compounds, for three runs in two optical channels. (b) Adjusted p for every test of descriptor distance against response distance: twelve exact Mantel cells and six Fisher combinations of the independent runs. (c) Cosine between the response pattern of DMMP and those of the four G agents and VX, by concentration, in the ultraviolet channel.

It is stable enough (Figure 1a). One compound's patterns at two concentrations correlate at +0.345 to +0.542, different compounds anti-correlate at −0.103 to −0.330, and the

ordering holds in all six run-channel cells. The prespecified test of that ordering, a sign-flip enumeration over the four to seven compounds of each cell, returns the smallest value its compound count allows in all six cells, from 0.0078 to 0.0625, so it reports a property of the design and not of the data, and we withdraw it. The same statistic is tested here against a null that permutes compound labels within each concentration, which leaves every centred vector untouched and changes only which response is assigned to which compound. All six cells clear $q = 0.05$. Four are sampled at 10,000 relabellings and return the smallest value that sampler can give, 9.999×10^{-5}, so the smallest adjusted value over the six, 0.00015, is an upper bound rather than an estimate; the two four-compound cells are exact, by enumeration of all 13,824 relabellings, and give one-sided values of 0.0046 and 0.0033. The censoring does not bear on the conclusion, because relabelling leaves a mean within-minus-between difference of +0.1024 to +0.3138 across the six cells against observed differences of +0.5578 to +0.7288, with Cohen's d_z over compounds from +0.92 to +5.12 as a descriptive effect size (Supporting Information Section S14).

Dye-wise centring drives the mean between-compound cosine negative by itself, so an earlier version's comparison against $-1/(n-1)$ does not hold and the permutation null is what quantifies the inflation instead. What it establishes is the per-compound statement and not the stronger pairwise one: in 13 of the 34 compound-and-cell instances some other compound's consensus pattern is closer to a compound than that compound's own agreement across concentrations (Supporting Information Sections S14 and S23).

The standard simulant responds opposite to three of the five agents measured with it

The array resolves the compounds, so the measurement can be put to the one compound the field uses in their place. DMMP is named here rather than treated as one of many, because the field treats it as the G-agent surrogate and because decontamination efficiency is quantified against it. The first run contains DMMP together with the four G agents and

VX, the only place in the three runs where a designated simulant was measured beside the agents it stands in for.

In the ultraviolet channel DMMP correlates with tabun at +0.764 and with sarin at +0.645, and anti-correlates with soman at −0.492, cyclosarin at −0.721 and VX at −0.702. In the reflectance channel the same comparison gives −0.112, +0.569, −0.479, −0.495 and −0.273. The mean over all four G agents is +0.049 in the ultraviolet channel and −0.129 in reflectance; over the three G agents other than tabun it is −0.189 and −0.135. Neither mean carries the claim on its own, because the sign changes with the agent and with the channel, and tabun changes sign between channels while sarin does not. What this measurement does not show is a compound that behaves like the G series as a class.

A weaker claim survives in one channel. If DMMP is G-like at all it should sit closer to the G series than to VX, and that contrast, against the mean over the three matched G agents and excluding tabun, is +0.513 in the ultraviolet channel and +0.137 in reflectance. The ranges quoted with them, +0.131 to +0.722 and −0.144 to +0.290, come from resampling the four design concentrations with replacement; four fixed design levels are not an exchangeable sample, so these are descriptive ranges and not tests, and the ultraviolet range lies above zero while the reflectance range spans it.

Concentration changes the picture (Figure 1c). The DMMP-to-sarin cosine rises monotonically from +0.363 at 10 μM to +0.771 at 500 μM, and the DMMP-to-cyclosarin cosine falls monotonically from −0.462 to −0.700 over the same range, with monotonicity perfect in three of the ten agent-channel series. That is also the strongest result the design can give: with four concentrations the two-sided exact permutation has a floor of $p = 0.083$, three series reach exactly that floor, and Benjamini–Hochberg correction over the ten leaves a smallest adjusted value of 0.278. We report the direction and not a significance claim.

What the direction supports is that the resemblance between this simulant and this agent is not constant over the range at which the array was characterised. Two limits sit on it, the solution-phase concentration axis and a signal-to-noise confound that these tables cannot

separate from it, and both are set out in Supporting Information Section S25.

Two further readings suggested themselves and did not survive their own controls, and we report both in Supporting Information Section S24 because the alternative is to present artefacts as results. One was a ranking over the ten agent-channel cells, which sits exactly on its own expectation; the other was a channel decomposition of the DMMP-to-tabun cosine, whose concentration onto two of the 29 dye channels is what the product of two centred unit vectors in 29 dimensions does rather than a property of this compound.

What remains after both controls is the set of measured values themselves, and one structural fact. The first run contains no candidate simulant other than DMMP, so the designation cannot be contrasted against an alternative in response space at all. The permutation null that the descriptor analysis of the next section relies on is, in response space, empty.

Descriptor distance does not resolve the difference

One compound is one compound. The practice that put DMMP there is general, and it rests on an implicit transfer: that a compound close to an agent in descriptor space will be close to it in whatever the detector measures. We tested that transfer directly, as the Mantel agreement between a descriptor-distance matrix and a response-distance matrix over the same compounds, referred to a null that enumerates every relabelling of the compounds, so the test is exact rather than sampled.

The test is run within each run and never across runs. The three runs share no compound, so the dye-wise sensitivity of one plate cannot be calibrated against another, and pooling across them would need a commutability assumption we cannot support. That restriction sets the power: six compounds in the first run and six in the second carry both a response pattern and a descriptor vector, giving fifteen pairs in each. The third run drops out because the source reports its four compounds as the codes A-230, A-232, A-234 and A-242, gives their structures as representative rather than assigned, and does not state which compound

each code denotes,[20] and nitrogen mustard drops out of the second run for the same reason.

Across the twelve cells, three spaces by two channels by two runs, nothing survives correction (Figure 1b). Three cells reach a nominal p below 0.05, all of them in the second run: the structural space in the ultraviolet channel at 0.0167, the merged space in the same channel at 0.0194, and the electronic space in the reflectance channel at 0.0444. Benjamini–Hochberg correction over the twelve leaves a smallest adjusted value of 0.1167. The pattern of the three is itself uninformative about transfer, because the two ultraviolet cells implicate the structural and merged spaces while the reflectance cell implicates the electronic space, and the same space does not reproduce across channels of the same plate.

Where those three cells sit matters more than their size. The second run contains seven authentic agents and no designated simulant, so whatever agreement they carry is agreement between one agent and another on one plate and not evidence that a simulant reproduces an agent. The only run containing a designated simulant beside the agents is the first, and it gives nothing in any space or channel, with $|\rho| \leq 0.225$ and every p at or above 0.229. The one run that could speak to simulant fidelity is silent, and the run that speaks is not about simulants.

Combining runs does not change the answer. The first and second runs share neither compounds nor plates, so their exact Mantel p values are independent and can be combined by Fisher's method, and we report all six combinations of three spaces and two channels rather than selecting one. The smallest adjusted value over the six is 0.2373, and none reaches 0.05 (Supporting Information Sections S15 and S12).

What this does not license is the claim that no transfer exists. Fifteen pairs is a weak test and the pooled rank correlations are correspondingly imprecise: the structural space in the ultraviolet channel gives $\rho = +0.31$ with a 95% interval from -0.45 to $+0.81$, excluding neither zero nor a strong association. The defensible statement is that the transfer on which simulant selection depends has not been established in the one body of authentic-agent response data where it could be tested, and that what would settle it is a run containing

a compound already measured in an existing run, which would fix the plate correction and open the 136 cross-run pairs that are closed today.

Descriptor space cannot supply the missing validation

The remaining question is whether descriptor space can at least establish the premise it is used for, that the incumbents sit closer to the agents than other compounds do. Designating three compounds out of a panel's non-agents is a choice among a finite set, so the incumbent designation can be referred to every alternative rather than to a sampled null: 2024 designations on the primary panel of thirty-five compounds, 680 on the reduced panel of twenty-eight.

On the primary panel the answer is flat (Figure 2). No agent of the eleven has a valid match, the smallest nominal p over the eighteen tests is 0.1917, and the Benjamini–Hochberg adjusted value is 0.6299.

What produces that zero is not chemical background but the second part of the criterion, a distance smaller than that to every compound outside the designated set. Sarin, soman and cyclosarin all sit inside the threshold of DMMP, at 1.1805, 1.4583 and 1.6041, and each fails because diisopropyl fluorophosphate is nearer still; tabun, outside the threshold at 2.5320, has diethyl cyanophosphonate nearer than DMMP. Those two are not interferents but compounds the sensing literature itself proposes as simulants. Narrowing the comparison set to the sixteen interferents, which is how the deposited distance table implements that part, lets the same three G agents pass and gives three rather than zero; we report the wider reading as primary because it is the criterion stated in Section , and both counts because the difference is not a rounding matter. What the panel does not support is the designation of the incumbents in particular, because better candidates exist in descriptor space.

The reduced panel, which omits seven compounds, five of them named as simulants in the literature and two of them pesticides, reads differently. Three agents acquire a valid match and the merged space returns a nominal one-sided p of 0.0250, with the structural space at

0.0279. Whether that survives depends on the correction: 0.1632 for Benjamini–Hochberg over the eighteen tests, 0.0772 within each space, 0.0794 for Westfall–Young, which uses the shared permutations and so handles the dependence exactly, and 0.0419 for the only scheme that fixes one statistic in advance. Significance here is a property of one correction scheme on one panel construction, and we report all four rather than the one that passes.

The surviving structural value is also an ordinary draw from the choices available to an analyst: random three-descriptor subspaces place it at the 22.4th percentile. One quantity does separate the incumbents, and it is easy to over-read: across 500 such subspaces the incumbent designation reaches $p < 0.05$ in 27.4% of them, seventh of 680 ($p = 0.0103$), which is a statement about the flexibility of the comparison and not about proximity to the agents (Supporting Information Sections S17, S18 and S19).

Panel composition decides the answer as well. Adding back the five compounds the sensing literature also treats as simulants moves the count from three valid matches to zero, because diisopropyl fluorophosphate is nearer to three G agents than any incumbent is and 2-chloroethyl ethyl sulfide is nearer to sulfur mustard than dichloromethane is. The incumbents do not move; what moves is whether anything closer is allowed into the comparison. Resampling the descriptor columns a thousand times says the same distributionally (Figure 2b): the incumbents return a median of zero valid matches, no match at all in 61.2% of replicates and at most one in 85.6%, against a median of five for the eight-compound set.

Three further freedoms move it further than the panel does, and all three are set out with their numbers in Supporting Information Section S11: the electronic-structure method, where recomputing the three electronic coordinates at B3LYP/def2-SVPD instead of GFN2-xTB takes that space from one valid match of eleven to five; the descriptor encoding, where dropping the four retained coordinates that are binary indicators carried by a single compound raises the count from three to four, with wider sweeps giving five and six; and the shape block, where dropping the seven derived moment-of-inertia coordinates takes it from three to zero. We kept the choice that produces the smallest count in each case, and report

the sweeps because that is the only evidence that we did not select a convention for its answer.

Five analyst choices therefore move the count of valid matches: the panel, the order in which the correlation filter is handed its columns, the conformer embedding seed, the descriptor encoding and the shape block. None of them moves the result of Section . Recomputing the twelve Mantel tests under the three encodings leaves the smallest Benjamini–Hochberg value at 0.1167, 0.4722 and 0.3889, and dropping the derived shape coordinates raises it to 0.1708 with the largest rank correlation unchanged at 0.571. A quantity that five arbitrary choices move between zero and six is not a measurement of anything, which is why we do not report a match count as a finding.

One descriptor the opening of this paper names first is absent from the panel above, and we tested it separately rather than argue around it. Volatility enters as one axis, $\log_{10}$ of measured vapour pressure, in two tests fixed before the values were seen and held outside the family of seventy-two. On pairing, the incumbent trio does beat chance, at a one-sided p of 0.0206 over all 680 designations, but by a margin of 0.0015 in standardised units, and the valid-match count returns one match at $p = 0.4647$. On agreement with measured response it is the one quantity in this paper that reaches $q = 0.0430$, in the reflectance channel, in a two-channel family we adopted after withdrawing the prespecified statistic for this test, so it is post-hoc and we say so. Read as one family those four numbers give 0.1389, and jointly with the twelve descriptor cells, over sixteen, 0.1556; nothing clears 0.05 under either reading (Supporting Information Section S16).

Neither the designation record nor a toxicity model fills the gap

Two places remain where the missing evidence might be found, and neither supplies it for the incumbents. We looked because the published criterion for choosing a low-toxicity simulant is physicochemical,[14] and such a criterion says by itself neither what measurement the compound was validated against nor how toxic it is.

The first is the standard-setting record. We traced each incumbent to the documents that designate it. DMMP is identified in a NIOSH programme for respirator materials resistance testing, where it is named a filtration simulant for GB and for HD and the properties that carry the choice are mass transfer and absorbent capacity.[3] The same document names carbon tetrachloride and cyclohexane for the same purpose, so DMMP is one of three rather than the designated compound. It carries a draft-for-discussion marking, which is worth saying plainly: the most-cited designation of the compound this field treats as the standard nerve agent simulant has never been issued as a final standard. Its stated reasons are not what a detection user would assume, and Supporting Information Section S8 quotes them: the volatility was chosen to sit between two agents rather than to match either, so a gap between DMMP and any single agent on that axis is the designation behaving as specified, and detectability entered as a property of the compound in an unspecified instrument rather than as a measured correspondence with an agent's response. DMMP is designated a second time, for a different agent and a different measurement, in a National Research Council assessment of the Man-in-Simulant Test programme, which lists it as the fabric-testing simulant for VX.[5] One compound therefore stands in for VX in one document and for GB and HD in another, against two measurements, neither of which is detection. DPGME appears in a US Army patent for a training composition, where the property required of it is detectability by ion mobility spectrometry.[4] For dichloromethane we found no designating document in the public record we searched; Supporting Information Section S8 gives the sources, the date and the three limits of that search, one of which is a Test Operations Procedure we identified but could not obtain. Each of the two traceable designations is a validation against a measurement. For DMMP those measurements are filter filtration and fabric penetration, neither of which is detection. For DPGME the measurement is detection, but by ion mobility spectrometry, a different transduction from the colorimetric response reanalysed here. A simulant is validated for a purpose, and that validation does not travel with the compound to a different purpose.

The second is regulatory toxicity (Figure 3). The axis on which the agents are placed here is the EPA acute exposure guideline level, and nineteen of the twenty-eight panel compounds carry one. Two of the three incumbents do not. DMMP, the compound against which decontamination efficiency is quantified, has no AEGL, and neither has DPGME; dichloromethane has one. Those two are not absent from every register: the US Department of Energy protective action criteria hold values for DMMP and NIOSH holds two for DPGME, tabulated with their sources in Supporting Information Section S27. What is missing is not a number of any kind but a number on the axis the agents occupy, because those values rest on different averaging times, severity definitions and derivations, so placing them beside a ten-minute AEGL-3 would compare quantities that are not the same quantity. A model cannot close the gap either. A random forest trained on 230 AEGL chemicals reaches $R^2 = 0.458$ in cross-validation with an out-of-fold bias of +0.001 log units, so nothing in its internal statistics warns a user off, and applied to twenty military agents held out of training it is wrong in one direction without exception, by 0.40 to 3.42 log units. Its applicability-domain check certifies fourteen of the twenty as inside the domain, and the worst of those fourteen is wrong by 2.68 log units, a factor of 482. The measured axis and its gaps, the effect size of that failure, the test of whether fidelity and acute toxicity are associated at all, which finds no monotonic relation across the compounds carrying both axes, and what the two axes together do and do not license are in Supporting Information Sections S28, S29 and S31.

Conclusions

The compound this field uses to stand in for the G agents responds opposite to three of the five agents it was measured beside, in both optical channels. That is one run and one array, and we report it as a measurement rather than as a test, but it is the measurement the practice has been missing. Simulant validity is not a property of a compound but a claim

about a measurement, and the practice of designating chemical warfare agent simulants records neither the measurement a compound was validated against nor the comparison in which its validity was tested. The criterion the literature does state is physicochemical alignment with the agent,[14] a property of the compound and not of the measurement.

Three counts fix where the evidence stands. In the three runs reanalysed here, one designated simulant was measured beside the agents, in one run of one array, and we found no second co-measurement in the literature we searched. Twelve exact Mantel tests and six combinations of them leave no agreement between descriptor distance and response distance below $q = 0.05$, against a design that resolves a rank correlation of 0.85 at 80% power at that corrected threshold, so that count bounds the design rather than the chemistry (Supporting Information Sections S14 and S21). On the primary panel no agent of the eleven has a valid match to an incumbent simulant, at a Benjamini–Hochberg adjusted value of 0.6299, and the compounds that displace the incumbents there are themselves literature-proposed simulants rather than interferents.

The documentary record accounts for the first of those counts rather than contradicting it. DMMP was designated against respirator filtration and fabric penetration and DPGME against detectability by ion mobility spectrometry, and a validation does not travel with a compound to a purpose it was never tested for. The toxicity axis stops in the same place: DMMP carries no AEGL, the values other registers hold for it are not the quantity the agents are placed on, and a random forest trained on 230 AEGL chemicals is wrong in one direction without exception on twenty held-out military agents. Absence of evidence is not evidence of unsuitability, so we conclude nothing about whether the incumbents are poor choices, and we propose no replacement compound either, because selecting one on descriptor similarity would repeat the error this paper is about.

Two readings this paper does not support should be ruled out, because the designating documents are cited here for what they measured. Filtration and surface decontamination are mass-transport and reactivity measurements, and nothing here bears on whether DMMP

is suitable for either; the same holds for DPGME and ion mobility spectrometry, which we did not measure. A compound can be a sound surrogate for one measurement and an unsound surrogate for another. The claim at issue is narrower: that a validation performed against one measurement is being carried, without a second validation, to colorimetric and spectroscopic detection of the agents themselves.

What can be specified is the measurement that would settle the question: a bridging run, with candidate simulants and the agents they stand in for exposed on the same plate, in the same optical channels, at the same concentrations, within one session. That yields the one quantity no amount of descriptor work can supply, a response-space distance measured under identical conditions, and it needs no new chemistry and no new instrument, only that the simulant be placed beside the agent rather than in place of it. Until that run exists, a statement that a compound is a validated detection simulant reports a convention, and a convention should be declared as one.

Experimental Section

Array measurements

The response data are the colorimetric tables of three published runs of one array, all three from this laboratory and sharing authors with the present paper, reanalysed here and not remeasured; the accuracy figures this paper withdraws and the agent provenance claim it corrects are our own. The first run covers tabun, sarin, soman, cyclosarin, VX and dimethyl methylphosphonate;[18] the second hydrogen cyanide, cyanogen chloride, phosgene, chloropicrin, sulfur mustard, nitrogen mustard and lewisite;[19] the third the Novichok agents A-230, A-232, A-234 and A-242.[20] We made no measurement of our own and handled no agent. Each source reports work at an Organisation for the Prohibition of Chemical Weapons designated laboratory under Chemical Weapons Convention procedures; the first names the CBRN Defense Research Institute, Seoul, as the source of the agents, and none of the three

says the agents were produced there. For the second run, purity was confirmed by GC–MS and exceeded 95% for every compound except lewisite, at 67%;[19] the first and third report no purity value, which the reader should treat as unstated rather than as high, and dropping lewisite changes no count reported here (Supporting Information Section S9).

The array is the same 29 commercial reagents in the same order in all three runs, used as received and listed in Supporting Information Table S3. Five are not chromogenic dyes: anthracene and pyrene are fluorophores, reduced L-glutathione is a thiol reducing agent, and 2,4-dinitrophenylhydrazine and 2-hydrazinobenzothiazole are carbonyl derivatising reagents, so part of the array responds by covalent derivatisation and part by solvatochromic shift. The sources agree on the identities and the order and disagree on one stock solvent, dimethyl sulfoxide for dye 15 in the first two runs and water in the third, which we report as published. Stock solutions at 100 mM in ethanol, methanol, dimethyl sulfoxide, water, tetrahydrofuran or toluene were diluted into acetonitrile to a working concentration of 100 μM, putting the residual stock solvent at 0.1% (v/v). Each analyte was delivered in acetonitrile by serial dilution to final concentrations of 0, 10, 50, 100 and 500 μM in a 2.0 mL dye solution, mixed without incubation and imaged at once under visible light and under ultraviolet excitation at 365 nm, with a smartphone camera in an enclosed box at 22 $\pm$ 3 °C and 35 $\pm$ 5% relative humidity. Regions of interest were central 20 $\times$ 10 pixel zones; RGB values were read at five positions per region, averaged, converted to CIE $L^*a^*b^*$, and expressed as the CIEDE2000 difference ΔE_{00} against the 0 μM image of the same dye in the same channel. The sources do not state the working colour space, the reference white or whether the camera's transfer function was linearised, so this work inherits those three choices rather than making them, and a signed readout would carry direction that ΔE_{00} discards. The three runs are different compounds on different plates and are not replicates, so no ΔE_{00} value is compared across runs; every analysis is within a run and channel, results are combined only at the level of correlation coefficients, and the tables hold one value per compound, concentration, dye and channel.

Response patterns

Each compound-concentration cell gives a 29-dye ΔE_{00} vector. The processing is three steps and the order matters: each vector is scaled to unit length, the compound-axis mean within that concentration is subtracted, and the result is scaled to unit length again. Agreement is then the cosine. The second scaling is what breaks the zero-sum property discussed in Supporting Information Section S14. Dye columns bit-identical across every row were counted once, which leaves 23 effective channels in the ultraviolet pivot of the second run and 25 in that of the third.

Descriptors

Structural and physicochemical descriptors, 51 after excluding run metadata, were computed with RDKit from canonical SMILES. Ten are read from a three-dimensional structure as Boltzmann-weighted means over an ETKDGv3 conformer ensemble at 298 K with MMFF94 energies, or UFF for lewisite 1, the one compound without MMFF parameters (Supporting Information Section S4). Electronic descriptors were computed with GFN2-xTB in the gas phase, Boltzmann averaged over up to five conformers at 298.15 K, on a separate embedding from the structural one, and the electronic space carries E_{HOMO}, E_{LUMO} and the dipole moment. The Parr electrophilicity index $\omega = \mu^2/2\eta$ and the condensed Fukui index at phosphorus are admitted to no distance, ω because it is an exact function of the two orbital energies that accompany it (Supporting Information Section S22). All three electronic coordinates were recomputed at B3LYP/def2-SVPD: the merged space keeps the same valid-match count and the same exact p under both methods while its individual distances move, and the three-coordinate electronic space changes answer (Section , Supporting Information Section S10). Identifiers and formulae were resolved against PubChem, and no property value was taken from a prediction service.

Distances and tests

Within each space the filter runs in three stages, in this order: constant columns are removed, then one of each pair correlated at $|r| > 0.95$, then any column that remains an exact algebraic function of others still present. The third stage overrides the protected set where the determining columns survive (Supporting Information Section S6). Remaining descriptors were standardised and distances taken as Euclidean, normalised by the median nearest-neighbour separation of the same space, which makes a normalised distance comparable within a panel and not between panels.

A paired comparison against the nearest interferent is confounded by group size, so we enumerated every way of designating three of the non-agents as simulants and report the rank of the actual designation as an exact p-value: $\binom{24}{3} = 2024$ designations on the primary panel and $\binom{17}{3} = 680$ on the twenty-eight-compound panel. The hypothesis is directional, so the reported p is one-sided; the two-sided values are given in Supporting Information Section S12, because on the twenty-eight-compound panel the convention decides the answer. The Mantel tests of Section are two-sided, since the direction of a matrix correlation is not fixed in advance. A valid match requires a normalised distance of 2 or less together with a distance smaller than that to every compound outside the designated set; that threshold is itself an analyst choice and Section reports what dropping it does. Sensitivity to descriptor choice was assessed by resampling the merged-space descriptors with replacement 1000 times under a fixed seed, restandardising and recomputing within each replicate, so those intervals are percentile ranges over descriptor choice and not confidence intervals.

Agreement between descriptor distance and response distance was tested by Mantel correlation with exact enumeration of relabellings in each run-channel cell, with the independent runs combined by Fisher's method and the family corrected by Benjamini–Hochberg. Six distance statistics and six multiplicity schemes were carried through in parallel so that the conclusion can be read against the analyst choices rather than from one.

Software

NumPy 2.5.3,[21] SciPy 1.18.1,[22] pandas 3.0.5,[23] scikit-learn 1.9.1,[24] Matplotlib 3.11.2,[25] RDKit 2026.03.6[26] and PySCF 2.14.0.[27] Conformer embedding used ETKDGv3,[28,29] the semiempirical electronic descriptors GFN2-xTB,[30,31] whose binary version was not recorded when the cache was built and is therefore not pinned, and the physicochemical descriptors the atomic-contribution and fragment methods implemented in RDKit.[32,33] All analyses are deterministic given the seed. The designation tests on both panels and the Mantel tests are exhaustive; the eight-simulant designation null and the random-subspace controls are sampled, at 20000 and 2000 draws from the recorded seed.

Data Availability

Analysis code, descriptor tables, the parsed regulatory values with their sources and every figure script are deposited in a public repository.[34] Every number we computed can be regenerated from it. Three classes are quoted rather than recomputed and are marked as such where they appear: values from the three source studies, values from regulatory registers, and the two cross-validated accuracies the deposited tables do not reproduce. The colorimetric input is the deposited ΔE_{00} table published with the source studies, and the conversion from plate images to those colour differences was performed there, outside this deposit.

Acknowledgement

The colorimetric response data reanalysed here were measured in three earlier studies from this laboratory,[18–20] and we thank the staff of the CBRN Defense Research Institute who performed those measurements under Chemical Weapons Convention procedures. K.K. carried out the conceptualisation, software, formal analysis and visualisation and wrote the original

draft. J.K. and J.Y. contributed to the analysis and validation, J.Y. to the original draft, M.K. to the validation and to reviewing and editing, D.-H.L. to validation, resources and supervision, and Y.J.J. to validation, supervision and project administration. This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. The authors declare no competing financial interest.

Supporting Information Available

The Supporting Information is available free of charge at `https://pubs.acs.org/doi/XX.XXXX/acssensors.XXXXXXX`.

- The acute inhalation toxicity axis and the gaps in it, with the predictive interpolation of that axis and the decoupling test against fidelity (Sections S3 and S26); conformer dependence of the shape descriptors; the density-functional check on the electronic coordinates; the full permutation family under all six multiplicity schemes; the dimensionality, subspace and block-weighting controls with their nulls; the five analyst freedoms in full; the measured array response and what it does and does not establish; phosphorus-local reactivity; and the structural filter (Section S30) (PDF)

References

(1) Askim, J. R.; Mahmoudi, M.; Suslick, K. S. Optical sensor arrays for chemical sensing: the optoelectronic nose. *Chemical Society Reviews* **2013**, *42*, 8649.

(2) Rakow, N. A.; Suslick, K. S. A colorimetric sensor array for odour visualization. *Nature* **2000**, *406*, 710–713.

(3) National Institute for Occupational Safety and Health *Chemical Warfare Agent (CWA) Simulant Program for Respirator Materials Resistance Testing*; 2002; Dated 31 July 2002.

(4) Seitzinger, A. T.; Genovese, J. A. Chemical Agent Simulant Training Composition. 2006; https://patents.google.com/patent/US7129094B1/en, U.S. Patent US 7,129,094 B1, assigned to the United States Department of the Army. Filed 16 May 2003; granted 31 October 2006.

(5) National Research Council *Technical Assessment of the Man-in-Simulant Test Program*; National Academies Press: Washington, DC, 1997; Chapter 3, Simulant Selection.

(6) ASTM International Standard Test Method for Man-In-Simulant Test (MIST) for Protective Ensembles. ASTM F2588-12(2020), 2020; Designation F2588-12, reapproved 2020.

(7) Lavoie, J.; Srinivasan, S.; Nagarajan, R. Using cheminformatics to find simulants for chemical warfare agents. *Journal of Hazardous Materials* **2011**, *194*, 85–91.

(8) Bartelt-Hunt, S. L.; Knappe, D. R. U.; Barlaz, M. A. A Review of Chemical Warfare Agent Simulants for the Study of Environmental Behavior. *Critical Reviews in Environmental Science and Technology* **2008**, *38*, 112–136.

(9) James, T.; Wyke, S.; Marczylo, T.; Collins, S.; Gaulton, T.; Foxall, K.; Amlôt, R.; Duarte-Davidson, R. Chemical warfare agent simulants for human volunteer trials of emergency decontamination: A systematic review. *Journal of Applied Toxicology* **2018**, *38*, 113–121.

(10) James, T.; Collins, S.; Marczylo, T. Identification of novel simulants for toxic industrial chemicals and chemical warfare agents for human decontamination studies: A systematic review and categorisation of physicochemical characteristics. *International Journal of Environmental Research and Public Health* **2021**, *18*, 8681.

(11) US Army Test and Evaluation Command *Test Operations Procedure (TOP) 08-2-196: Simulant Selection for Laboratory, Chamber, and Field Testing*; 2011; Known to this

work only through the citation in TOP 08-2-188 CN1, which is also the source of the title and year. A revision 08-2-196A is held by DTIC under accession AD1167884. Neither was obtainable for this work.

(12) Agrawal, M.; Sava Gallis, D. F.; Greathouse, J. A.; Sholl, D. S. How Useful Are Common Simulants of Chemical Warfare Agents at Predicting Adsorption Behavior? *The Journal of Physical Chemistry C* **2018**, *122*, 26061–26069.

(13) Lawrence, D. S.; Van Houten, K. A. Simulants for Chemical and Explosive Threats. *Johns Hopkins APL Technical Digest* **2025**, *37*.

(14) Webb, S.; Coulon, F.; Temple, T. A critical review of liquid, low toxicity chemical warfare agent simulants: Enhancing accuracy, safety, and methodological approaches for sampling. *Journal of Hazardous Materials* **2025**, *492*, 138021.

(15) Chen, Y.; Liu, J. A Practical Guide of Choosing and Validating Aptamers for Biosensors. *ACS Sensors* **2026**, Advance online publication.

(16) Yoo, J.; Lee, D.; Lee, S.; Kang, S.; Kim, H. I.; Jang, Y. J.; Kim, J.; Park, T. H. Combinatorial Pattern Response of Bioelectronic Nose for the Detection of Real Nerve Agents. *ACS Sensors* **2025**, *10*, 185–195.

(17) Bae, S.; Kang, K.; Kim, Y. K.; Jang, Y. J.; Lee, D.-H. Field-deployable real-time AI System for chemical warfare agent detection using YOLOv8 and colorimetric sensors. *Chemometrics and Intelligent Laboratory Systems* **2025**, *261*, 105365.

(18) Kim, J.; Kang, K.; Shin, M.; Kim, S.; Yoo, J.; Kim, M.-K.; Lee, W. B.; Lee, D.-H. Machine learning-enhanced colorimetric sensor array for rapid detection of nerve agents. *Journal of Hazardous Materials* **2026**, *501*, 140703.

(19) Kim, S.; Yoo, J.; Kang, K.; Kim, J.; Shin, M.; Cha, Y. K.; Churchill, D. G.; Kim, M.-K.;

Lee, D.-H. A multiplex chromogenic–fluorogenic sensor array for comprehensive optical discrimination of chemical warfare agents. *Discover Sensors* **2025**, *1*, 29.

(20) Shin, M.; Kang, K.; Kim, J.; Kim, S.; Yoo, J.; Jang, Y. J.; Kim, M.-K.; Anslyn, E. V.; Lee, D.-H. Portable machine learning–integrated optical sensor array for rapid and selective detection of authentic Novichok nerve agents. *Sensors and Actuators B: Chemical* **2026**, *457*, 139706.

(21) Harris, C. R. et al. Array programming with NumPy. *Nature* **2020**, *585*, 357–362.

(22) Virtanen, P. et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* **2020**, *17*, 261–272.

(23) McKinney, W. *Data Structures for Statistical Computing in Python*; 2010; pp 56–61.

(24) Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* **2011**, *12*, 2825–2830.

(25) Hunter, J. D. Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering* **2007**, *9*, 90–95.

(26) Landrum, G.; others RDKit: Open-source cheminformatics. `https://www.rdkit.org`, 2026; Version 2026.03.6. Cited by name, version and project URL: the Zenodo DOIs registered for RDKit identify individual releases (10.5281/zenodo.10398 is release 2014.03.1) and no version-independent concept DOI could be verified, so none is given here.

(27) Sun, Q.; Berkelbach, T. C.; Blunt, N. S.; Booth, G. H.; Guo, S.; Li, Z.; Liu, J.; McClain, J. D.; Sayfutyarova, E. R.; Sharma, S.; Wouters, S.; Chan, G. K.-L. PySCF: the Python-based simulations of chemistry framework. *WIREs Computational Molecular Science* **2018**, *8*, e1340.

(28) Riniker, S.; Landrum, G. A. Better Informed Distance Geometry: Using What We Know To Improve Conformation Generation. *Journal of Chemical Information and Modeling* **2015**, *55*, 2562–2574.

(29) Wang, S.; Witek, J.; Landrum, G. A.; Riniker, S. Improving Conformer Generation for Small Rings and Macrocycles Based on Distance Geometry and Experimental Torsional-Angle Preferences. *Journal of Chemical Information and Modeling* **2020**, *60*, 2044–2058.

(30) Grimme, S.; Bannwarth, C.; Shushkov, P. A Robust and Accurate Tight-Binding Quantum Chemical Method for Structures, Vibrational Frequencies, and Noncovalent Interactions of Large Molecular Systems Parametrized for All spd-Block Elements (Z = 1–86). *Journal of Chemical Theory and Computation* **2017**, *13*, 1989–2009.

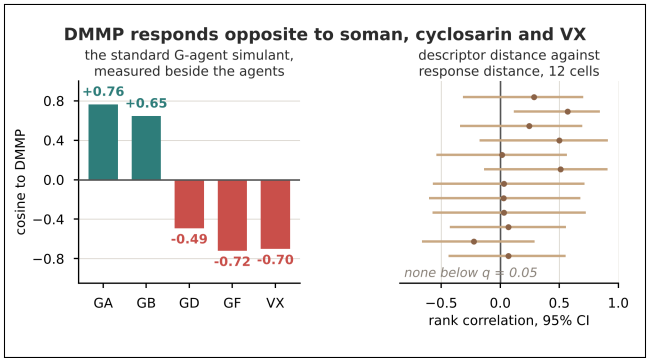
(31) Bannwarth, C.; Ehlert, S.; Grimme, S. GFN2-xTB—An Accurate and Broadly Parametrized Self-Consistent Tight-Binding Quantum Chemical Method with Multipole Electrostatics and Density-Dependent Dispersion Contributions. *Journal of Chemical Theory and Computation* **2019**, *15*, 1652–1671.

(32) Wildman, S. A.; Crippen, G. M. Prediction of Physicochemical Parameters by Atomic Contributions. *Journal of Chemical Information and Computer Sciences* **1999**, *39*, 868–873.

(33) Ertl, P.; Rohde, B.; Selzer, P. Fast Calculation of Molecular Polar Surface Area as a Sum of Fragment-Based Contributions and Its Application to the Prediction of Drug Transport Properties. *Journal of Medicinal Chemistry* **2000**, *43*, 3714–3717.

(34) Kang, K. simulant-designation-evidence: analysis code, descriptor tables and figure scripts for this work. `https://github.com/bisu9082/simulant-designation-evidence`, 2026; Public Git repository.

TOC Graphic


DMMP responds opposite to soman, cyclosarin and VX
the standard G-agent simulant, measured beside the agents
descriptor distance against response distance, 12 cells
cosine to DMMP
0.8
0.4
0.0
−0.4
−0.8
+0.76
+0.65
-0.49
-0.72
-0.70
GA
GB
GD
GF
VX
none below q = 0.05
−0.5
0.0
0.5
1.0
rank correlation, 95% CI

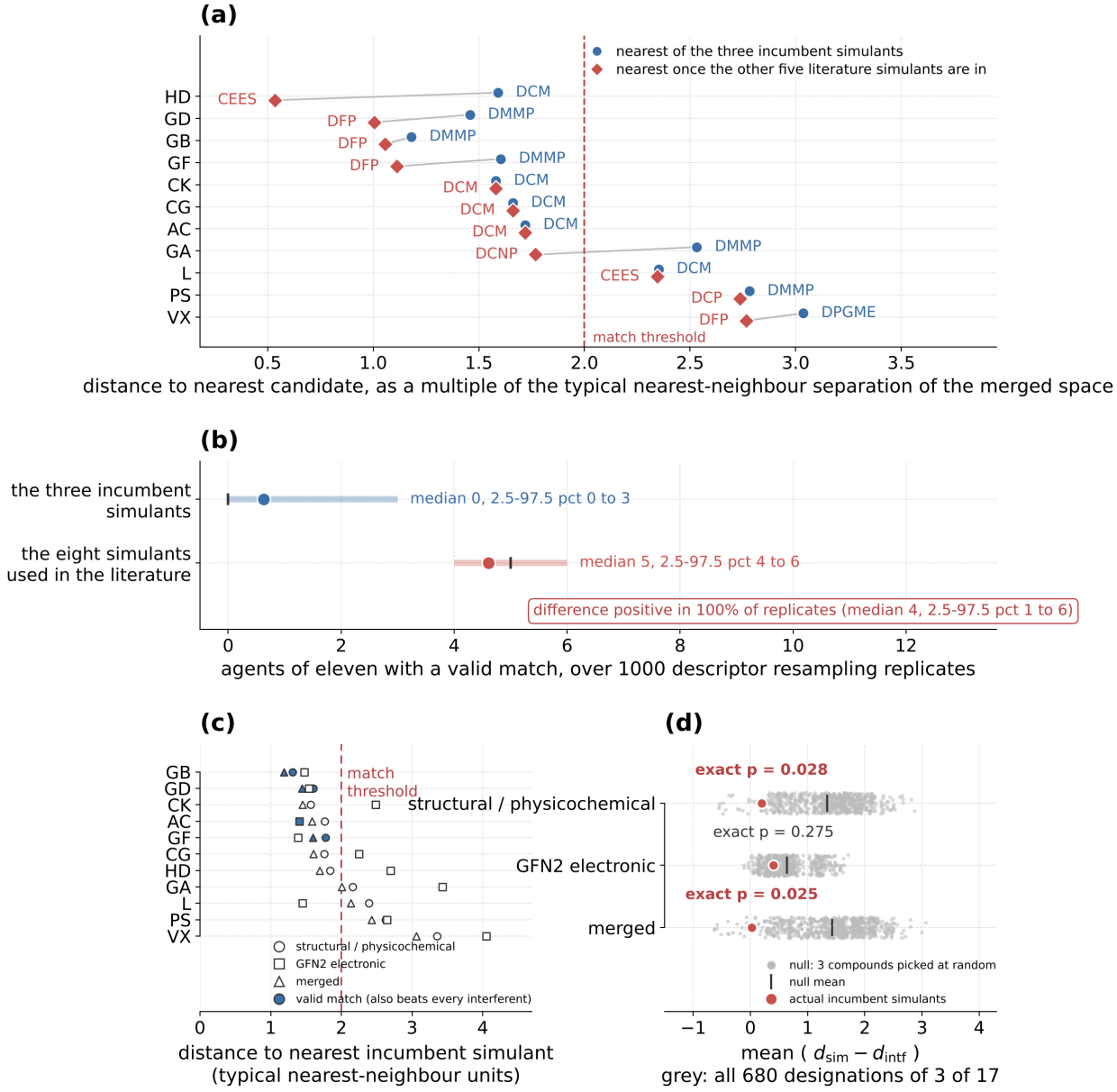


Figure 2: Descriptor-space tests. (a) Distance from each agent to its nearest candidate, for the three incumbents and for the eight compounds the sensing literature also calls simulants. (b) Valid matches over 1000 descriptor resampling replicates. (c) Distance to the nearest incumbent by space. (d) Observed mean difference against the exact null over all 680 designations of three compounds from seventeen.

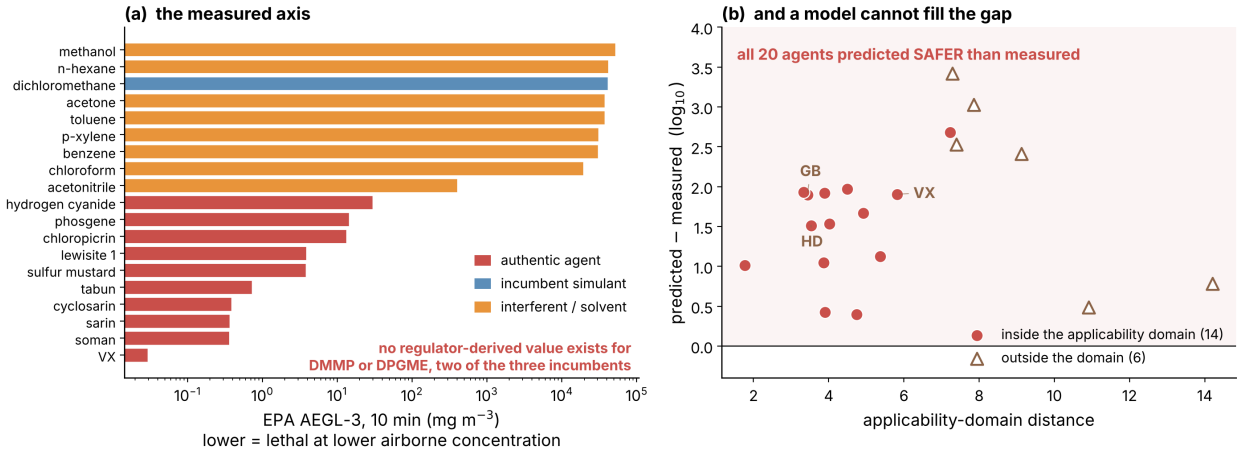


Figure 3: The toxicity axis. (a) Measured EPA AEGL-3 10-minute values for the panel compounds that carry one. (b) Prediction error of a random forest trained on 230 AEGL chemicals, for twenty military agents held out of training, against distance to the training set.